SE301 Project 2026-27

Singapore Vertical Farming Research Management System (SVFRMS)
Refactoring & Modernisation: Technical Report

Group: 6

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1. Legacy System Analysis

SVFRMS monitors the growing chambers of a vertical-farming facility. Each chamber grows one crop. Sensors report temperature, humidity, soil moisture, light and nutrient levels. A reading outside the crop's range raises an alert, records water or energy use and triggers a corrective device (cooling fan, water pump, humidifier, …). At the end of a run the system archives alerts, prints a daily report and saves an archive report to `archive/daily_report.txt`.

The legacy code is one 393-line class, `SVFRMS`, whose main components are:

Component	Legacy location
Global mutable state: alert/log/report lists, counters, chamber maps, four locks	`public static` fields, lines 7–31
Chamber state and per-chamber summary	nested classes `ChamberState`, `ChamberSummary`, lines 33–60
Event handling: logging, alert creation, tracking	`handleEvent(String type, …)` string switch, lines 62–88
Device control and its queue (producer–consumer)	`activateControl` if/else chain (90–129); `addControlRequest` + `ControlThread`, `wait/notifyAll` on an `ArrayList` (131–139, 251–273)
Crop threshold rules	`processSensor`: 65 lines of nested `if (crop.equals(...))`, lines 141–205
Reports	`generateDailyReport`, `archiveAlerts`, `saveArchiveReport`, lines 207–241, 309–347
Simulation and start-up	`ChamberThread extends Thread` (275–295), `main` (349–392)

2. Code Analysis and Limitations

2.1 Design issues and code smells

#	Issue (code smell)	Evidence	Principle
D1	**God class** (large class, divergent change): sensing, crop rules, devices, alerts, reports, file I/O, threads and simulation in one class.	whole file	SRP
D2	**Crop rules hard-coded** (long method, duplicated code, shotgun surgery): adding Kale means a new `if` block in up to three sensor branches of `processSensor`; thresholds, costs and devices are **magic numbers** repeated per crop.	lines 145–188	OCP
D3	**Stringly-typed dispatch** (primitive obsession): `handleEvent("TRACK_ALERT", …)`, `activateControl(…, "Cooling Fan")`; queue items are `chamber + "|" + control`, split again later. Typos fail silently; a `|` in a chamber name breaks parsing.	lines 62–129, 136, 269	OCP, type safety
D4	**Inheritance where composition belongs.** `ChamberThread` and `ControlThread` extend `Thread` to get a thread for their `run()`; they are tasks, not specialised threads. No LSP violation was found, because the code has no type hierarchy at all: nothing is ever substituted, so nothing *can* be substituted, and no new behaviour can be plugged in without editing the class. Extending `Thread` also fixes one thread per task, and `ControlThread` swallows every exception around `wait()`, so `interrupt()` cannot stop it (D8).	lines 251–273, 258, 275	LSP (no substitutable types), composition over inheritance
D5	**Fat, implicit interface**: `handleEvent` is one entry point for six unrelated operations; every caller depends on all of them.	line 62	ISP
D6	**Dependence on globals and concretions**: every collaborator is a `public static` field; nothing can be replaced or tested alone.	lines 10–31	DIP
D7	**Dead code**: `waterPumpLock`, `nutrientLock`, `irrigationCycle`, `feedingCycle`, `showAlerts`, `showLogs`, the `ALERT_LOG` branch, commented-out `sleep` calls — none is ever called.	lines 28–29, 66–68, 224, 243–249, 297–307, 385	Clean code
D8	**Swallowed exceptions**: `catch(Exception e){}` hides interrupts; `saveArchiveReport` prints an I/O error and the run still "succeeds".	lines 134, 258, 344	Reliability

2.2 Behavioural quirks (kept, see §3.3)

- Soil checks exist only for Tomato (and only its minimum); humidity checks only for Tomato and Strawberry. The spec's soil range for Strawberry and humidity/soil ranges for Lettuce were never coded.
- Light (< 500 lux) and nutrient (< 40 %) limits are facility-wide, and their alert text omits the crop. Low temperature alerts but starts no device (no Heating Unit).
- A queued device request can run after a newer reading of the same sensor (a fan queued at 35 °C runs after a 20 °C reading). This is a logical ordering issue, not a data race.
- Never implemented: severity levels, Sensor Failure alerts, Heating Unit, Ventilation System, Shade Controller and the weekly, water, energy and nutrient reports. §5 shows where each plugs in; severity and Sensor Failure were added (§3.3).

2.3 Concurrency issues

Four `ChamberThreads` and one `ControlThread` share all static state with no consistent locking.

#	Type	Where	Failure
C1	Race condition	`alerts`, `sensorLogs`, `reports` are plain `ArrayLists` written by 4 threads	Lost alerts, corrupted array, `ArrayIndexOutOfBoundsException`
C2	Check-then-act race	`getChamberState`/`getChamberSummary`: `containsKey` → `put` on a `HashMap`	Two states for one chamber; `HashMap` corruption → `NullPointerException`
C3	Lost update	`totalWaterUsage += …`, `totalEnergyUsage`, `totalAlerts++`, `totalAlertsGenerated++`	Non-atomic read-modify-write → wrong totals
C4	Data race / visibility	`ChamberState` fields written by chamber threads and read-modify-written by the control thread with no lock	Device effect lost or computed from a stale value
C5	Lost update	`ChamberSummary.alerts++` (chamber threads) vs `mitigationActions++` (control thread)	Summary counts don't add up
C6	Order race	`main` starts `archiveAlerts` and `generateDailyReport` threads at once	"Active Alerts" is 0 or N at random, and the archive's activity lines ("Daily Report", "Archived N alerts") come in either order (the commented `sleep(20)` shows it was noticed)
C7	Liveness (deadlock-like hang)	Producers `wait()` on a full queue with no timeout; if `ControlThread` dies (any `RuntimeException`, e.g. the C2 NPE) nobody consumes	All chamber threads block forever; `main` hangs in `join()`
C8	Latent deadlock	Nested `synchronized(alertLock){ synchronized(reportLock){…} }` in two methods (lines 208–209, 223–230); the locks are `public static`, **non-final**	No deadlock today, because both methods take the locks in the same order (alert, then report), and `wait()` releases the queue monitor. But any new code that takes them in the other order, or reassigns a lock, deadlocks or breaks mutual exclusion
C9	Unsafe iteration	`showAlerts`/`showLogs` iterate lists other threads modify	`ConcurrentModificationException`

Measured (150 runs per version, Linux, JDK 21; each run checked against its own synchronized console output). The rates depend on the platform: on Windows 11 with JDK 25 the legacy version crashed with a `NullPointerException` in 42 of 60 runs and the refactored version in none; `tools/compare_legacy.sh` reproduces the count and the output comparison of §3.3.

Symptom	Legacy	Refactored
Crash: `NullPointerException` in a chamber thread (C2)	3	0
Archive lists fewer alerts than were raised (C1)	5	0
Wrong water/energy total (C3) / wrong "Generated Alerts" (C3)	5 / 2	0 / 0
Chamber summary counts wrong (C5)	1	0
"Active Alerts" in the daily report (C6)	0 in 130 runs, N in 20	0 in all 150

No hang (C7) occurred in these runs because the NPEs hit chamber threads. `legacy/LegacyHangDemo.java` forces the same NPE in the control thread, by swapping the public, non-final `chamberStates` field (D6) without editing the legacy code. It hangs in 5 of 5 runs: all four chamber threads wait forever in `addControlRequest` (line 134) and `main` never returns. Refactored, a dead worker wakes and rejects blocked producers (`ControlDispatcherLifecycleTest`), and `SVFRMS.run` turns the failed run into exit status 1 with no report (`SimulationRunnerTest`).

3. Proposed Design

3.1 Key decisions, by SOLID principle

Principle	Change (fixes)
SRP (D1)	Nine packages plus the composition root, one reason to change each (§4). The old dispatcher was split again: `ControlDispatcher` owns only the worker's lifecycle, `ClosableBoundedQueue` only blocking, `RequestExecutor` only running a request, `Listeners` only fault-isolated notification.
OCP (D2, D3)	Crops are data: a `CropProfile` holds a `Range` per sensor and the cost per breach. `SensorProcessor` has one generic rule (*breach → alert → cost → device*); no crop name appears in logic. Crops can even be added or removed at run time with no rebuild (`--add-crop`, `--crops=FILE`, `--remove-crop`). Every "for each sensor" loop iterates `SensorType`.
LSP (D4)	Composition replaces `Thread` inheritance: chamber work is a `Callable` run by an `ExecutorService`, and the dispatcher owns its worker. The new design has real substitution points, and the tests use them: scripted `ReadingSources` (`SensorExtensibilityTest`), throwing and blocking `Devices` (`ControlDispatcherLifecycleTest`), lambda `DeviceControllers` and failing listeners (`ListenerIsolationTest`). Each stand-in keeps the interface's documented contract, so the code under test cannot tell it apart from the production class.
ISP (D5)	Small, client-specific roles (`AlertListener`, `DeviceListener`, `DeviceController`, `Device`, `ChamberControl`, `Report`, `ReadingSource`): each client depends only on what it uses. Services expose a separate interface per client: `AlertService` is an `AlertSink` to monitoring and an `AlertLedger` to reporting; `ResourceMeter` is a `UsageRecorder` and `UsageTotals`; `ChamberStates` lets monitoring only `record` a reading and gives devices only a `ChamberControl`.
DIP (D6)	Constructor injection: the composition root (`SVFRMS.run`, `CommandLine`) chooses and creates every replaceable collaborator, including the `Reports` handed to `ReportService`. Services build only their own private parts (the dispatcher's queue, worker thread and executor; a listener list). `ArchitectureTest` checks by reflection that every constructor of `SensorProcessor`, `ReportService`, `ChamberSimulation`, `ControlDispatcher`, `RequestExecutor` and `AlertService` takes only interfaces or immutable values.

Dead code was removed (D7), including the sensor log that legacy wrote but never read; no production method exists only for tests. I/O errors now set the exit status, and interrupts are restored (D8).

3.2 Concurrency fixes

Issue	Fix
C1, C9	Each shared list has one owner that guards it (`AlertService` synchronized; the activity log a `synchronizedList`). Readers get immutable copies, so iteration never races.
C2	`ConcurrentHashMap.computeIfAbsent` (atomic get-or-create) in `ChamberRegistry` and `ChamberSummaryTracker`.
C3, C5	`AtomicInteger` counters in `ResourceMeter`; alert counters change under the same lock as the alert list; per-chamber summaries are `synchronized`.
C4	`ChamberState` is an `EnumMap` behind `synchronized` methods; every device change is one atomic `adjust` (read, add, clamp, write), so no effect is lost. Device code runs outside the lock (`deviceCodeRunsWithNoChamberLockHeld`). The ordering quirk of §2.2 is functionality, not a race, so it is kept.
C6	Archive, then daily report, sequentially, after all producers and the consumer have finished.
C7	`ClosableBoundedQueue` (one `ReentrantLock`, two `Conditions`) replaces `wait/notify`. If the worker dies, `abort` wakes every blocked producer at once with an error: no hang and no polling. Shutdown takes effect when the queue closes (admission is decided under the queue's lock): a request admitted before then runs, one after is rejected, also when `close()` comes before `start()`. `close()` lets the worker drain the queue, waits at most 30 s and reports any request not executed; a second concurrent `close()` waits for the same drain. Devices that throw are logged and counted; the worker carries on. Chamber tasks are `Callables` and `SimulationRunner` checks every `Future`, so a failed chamber fails the run (exit 1); an interrupted runner cancels its chambers instead of waiting for them.
C8	**No nested locks anywhere**, by construction: each component guards its own state with one lock, and listeners and device code run outside every lock. Reports read an immutable `FacilitySnapshot`, built after every thread has finished.

Tests: `countersSurviveConcurrentUpdates` (8 threads × 5 000 updates, exact totals), `closeDuringHeavyProductionLosesNothing` (close races 8 producers; every request runs or is rejected), `deviceCodeRunsWithNoChamberLockHeld`, worker death with a full queue, a hanging device, close before start, two concurrent closes, an interrupted runner, a sub-millisecond shutdown timeout, and a failed run ending in exit 1 with no report.

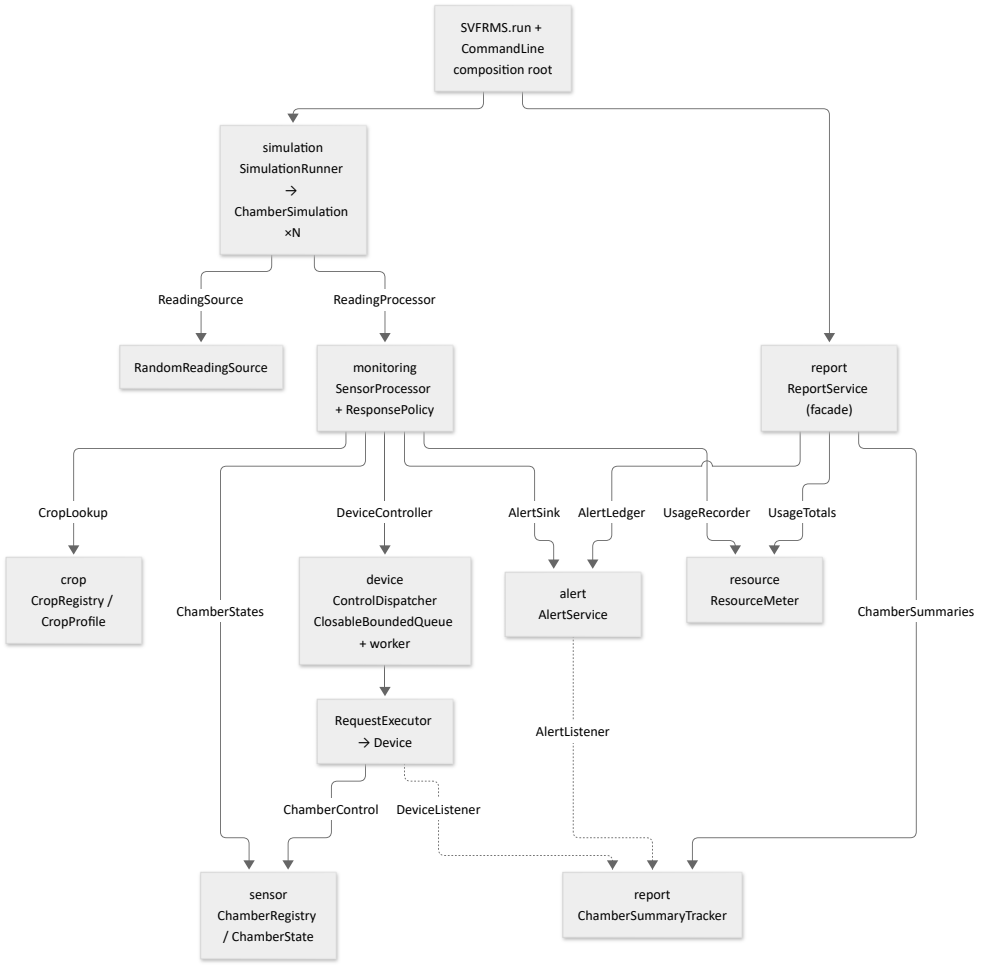
3.3 Behaviour preservation

Thresholds, alert texts, resource costs, device effects, console formats and the archive layout are identical: the sets of distinct output lines of 25 legacy and 25 refactored runs (numbers masked) were diffed and match (`tools/compare_legacy.sh`). The quirks in §2.2 are kept because the brief forbids changing functionality; each is now a one-line data change (e.g. `.soilMoisture(50, 65)` on Strawberry). The one bound legacy ignores, Tomato soil above 60 %, is declared on the crop (`.ignoreBreach(SOIL_MOISTURE, HIGH)`) rather than hidden in the policy, which must cover every sensor and direction. Deliberate changes appear only when triggered:

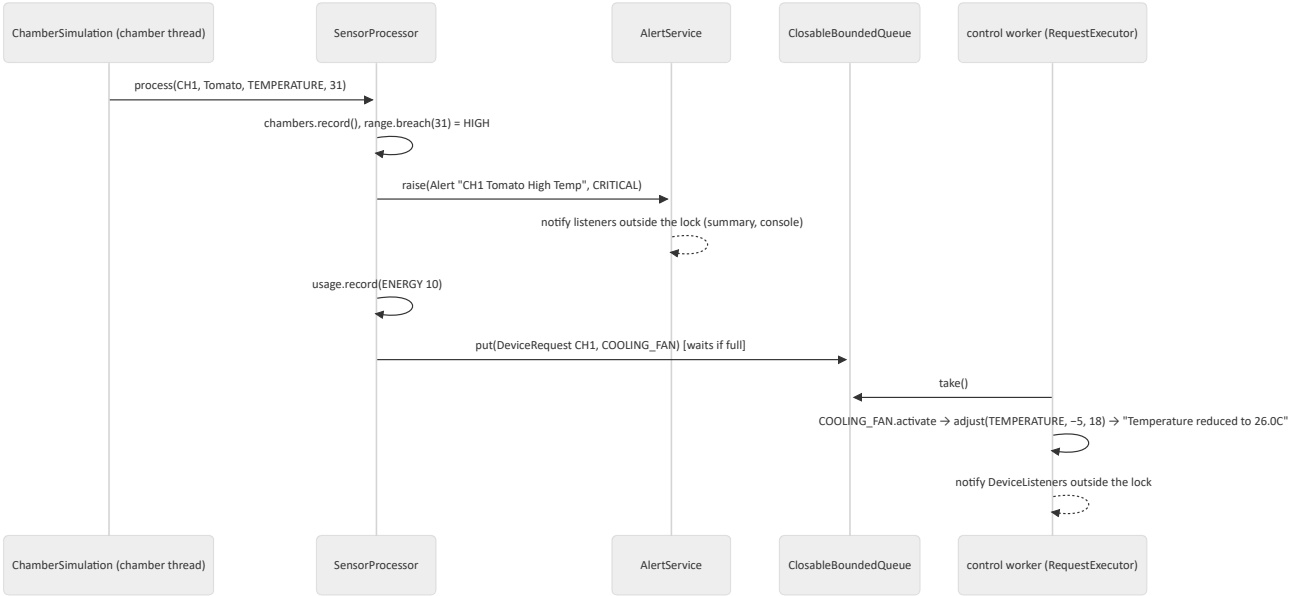
- C6 is deterministic: "Active Alerts" is always 0, the legacy majority outcome.
- **Sensor Failure** (spec §3.4): a reading a sensor cannot produce (NaN, infinite, outside its physical range, e.g. 150 % humidity) or no reading at all raises a CRITICAL `CHn Sensor Failure: …` alert and is never stored or acted on. Ranges, costs and resource amounts are validated.
- **Severity** (spec §3.4): every alert carries INFO, WARNING or CRITICAL (temperature breaches are CRITICAL, as in the spec's example). `--alert-format=severity` prints `CRITICAL: CH3 Lettuce High Temp`; the default console format is unchanged.
- `CHn=Crop` arguments replace the default four chambers (`java -jar run.jar CH1=Tomato CH5=Kale` runs two). New crops get alerts on both bounds of every range they define (e.g. `CH5 Kale High Soil Moisture`).
- If the archive cannot be saved, the legacy message is printed and the cause goes to stderr.
- Exit status: 0 success; 1 run failed (no report written); 2 archive not saved; 3 invalid arguments.

4. Final Architecture

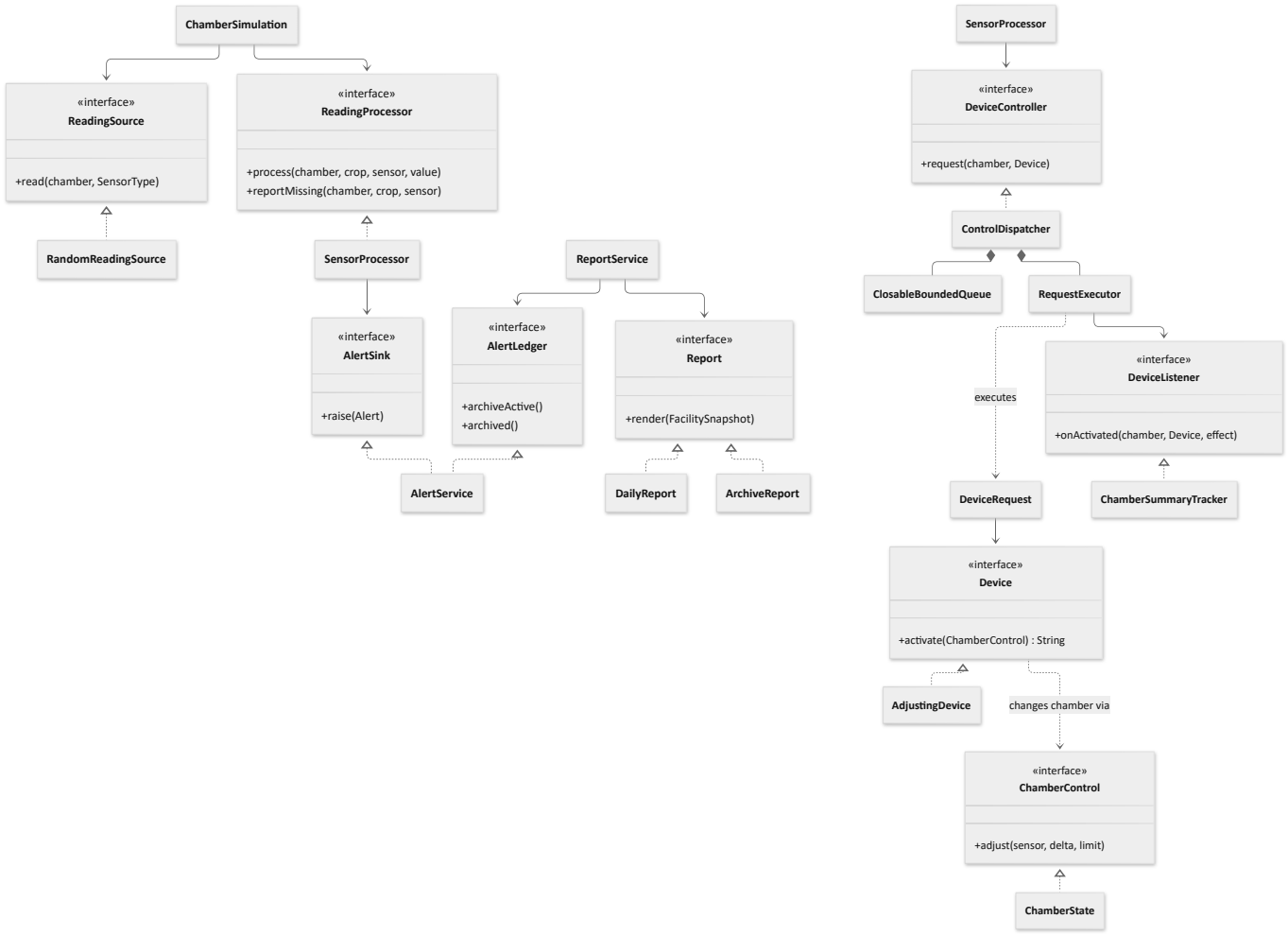
Component diagram



Sequence: one out-of-range reading



Class diagrams: monitoring, alerts and reporting (left); device control (right)



Package	Main classes	Responsibility
`crop`	`CropProfile` (+Builder), `Range`, `CropRegistry`, `CropCatalog`, `CropSpecParser`	What each crop needs
`sensor`	`SensorType`, `Breach`, `ChamberState`, `ChamberRegistry`	Readings and chamber conditions
`monitoring`	`SensorProcessor`, `ResponsePolicy`, `Response`	Reading → alert / cost / device rule
`device`	`ControlDispatcher`, `ClosableBoundedQueue`, `RequestExecutor`, `DeviceRequest`, `Devices`	Executing corrective actions
`alert`, `resource`	`AlertService`, `Severity`, printers; `ResourceMeter`, `Usage`	Alert lifecycle; water and energy
`report`	`ReportService`, `DailyReport`, `ArchiveReport`, `FacilitySnapshot`, `ChamberSummaryTracker`	Reporting
`simulation`, `support`	`ChamberSimulation`, `SimulationRunner`, `RandomReadingSource`; `Listeners`	Sensor feed; isolated notification

5. Extensibility and Maintainability

Change	What to touch
Add / remove a crop	One `CropCatalog` entry, or no code at all: `--add-crop="Okra:temperature=24..30,soil=50..70"`, `--crops=FILE`, `--remove-crop=Tomato`.
Add a device (e.g. Heating Unit)	One `Devices` constant, referenced from the `Low Temp` entry of `ResponsePolicy.standard()`; no logic changes.
New sensor type	One `SensorType` constant, its two `ResponsePolicy` entries and its simulated range; start-up fails fast if either is missing. Parsing, simulation and state follow automatically.
New report (weekly, water, energy, nutrient)	A new `Report` implementation rendering `FacilitySnapshot`.
New alert channel or severity routing	A new `AlertListener` registered in `SVFRMS.run`; a faulty one cannot disturb the others.
Real sensors	A new `ReadingSource`; an unresponsive sensor already raises a Sensor Failure alert.

Every change is additive and local. Constructor injection lets each component be tested alone: 68 tests in 14 classes cover behaviour parity, every crop bound, validation, concurrency, shutdown, run-time crops and exit codes.

6. Design Patterns

Pattern	Where	Problem solved	Why this rather than the alternative
Builder	`CropProfile.Builder`	A crop has up to five optional ranges, costs and exceptions, plus facility defaults.	A constructor with a dozen optional arguments is unreadable and error-prone; setters would make profiles mutable while chamber threads read them. The builder validates each range and yields an immutable profile.
Strategy	`Report` (`DailyReport`, `ArchiveReport`, injected into `ReportService`); `ReadingSource` (`RandomReadingSource`, test sources, future hardware)	Interchangeable algorithms chosen by the composition root.	The legacy approach, `if/else` on a type string, edits the caller for every new variant (D3).
Observer	`AlertService` → `AlertListeners`; `RequestExecutor` → `DeviceListeners`	Producers must not know who consumes alerts (console, summary, future email).	Direct calls (`handleEvent`) couple alerting to every consumer; fault isolation in `Listeners` keeps one bad listener from stopping the rest.
Command	`DeviceRequest` (chamber + `Device`; `Device.activate` is the action, `RequestExecutor` the invoker)	Deciding to act (chamber thread) is separated from acting (control thread); the request carries everything needed to run later, in order.	Legacy encoded commands as `"chamber|control"` strings parsed later (D3).
Producer–Consumer with guarded suspension	`ControlDispatcher` + `ClosableBoundedQueue`	Back-pressure, one writer for device effects, clean and abnormal shutdown.	`ArrayBlockingQueue` + poison pill (our first version) cannot wake producers when the consumer dies. A bounded `ThreadPoolExecutor` rejects new work when full instead of blocking the producer, so it loses legacy back-pressure.
Facade	`ReportService`	End-of-run reporting coordinates alerts, usage, summaries, two reports and file I/O.	Callers make three calls instead of reaching into five components, as legacy `main` did.
Null Object	`SensorProcessor`: an unknown crop gets an empty `CropProfile` (facility defaults only)	Unknown crops still get the facility-wide light and nutrient checks, as in legacy.	`null` checks at every use of the profile.

Supporting idioms: **Static factory methods** (`CropCatalog.defaultRegistry()`, `ResponsePolicy.standard()`, `RandomReadingSource.legacy()`) name the legacy configurations; **Registry** (`CropRegistry`, `ChamberRegistry`, thread-safe lookup); **Immutable Value Objects** (`FacilitySnapshot`, `CropProfile`, records), so reports never hold a lock; **Dependency Injection** with a single composition root (`SVFRMS.run`).

7. Build and Run

`mvn clean package` builds `target/run.jar` and runs the 68 tests. `java -jar run.jar` runs the legacy four chambers; `java -jar run.jar --help` lists the options.